

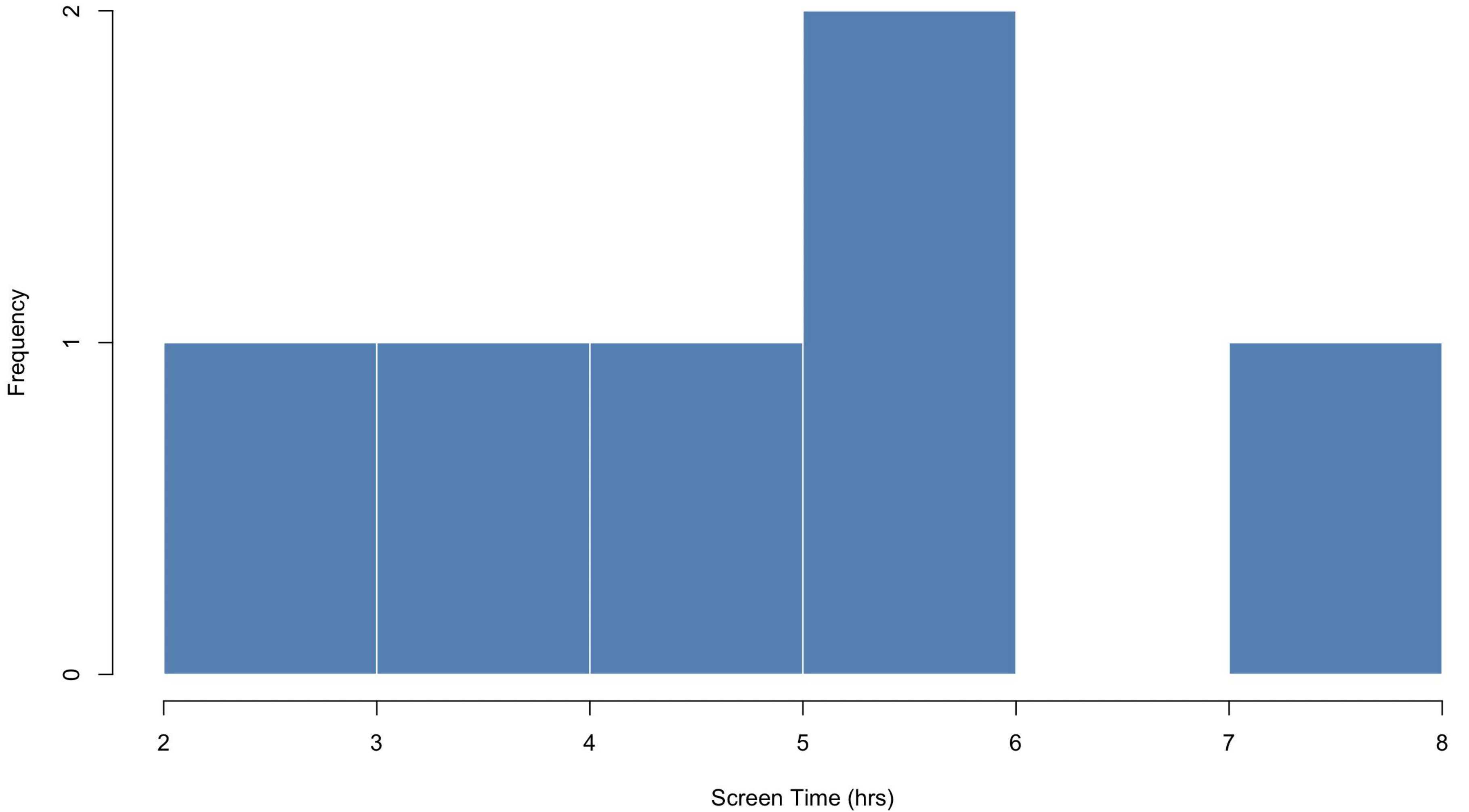


```
6      U06 Female  20      5.4      22      3.1      4
  Usage_Date
1 2025-01-08
2 2025-01-08
3 2025-02-11
4 2025-02-11
5 2025-03-14
6 2025-03-14
>
> library(ggplot2)
>
> new_window <- function() {
+   if (.Platform$OS.type == "windows") {
+     windows()
+   } else {
+     quartz()
+   }
+ }
>
> # -----
> # Q1a: Histogram - Screen Time
> # -----
> new_window()
> hist(app_data$Screen_Time,
+       col = "steelblue",
+       main = "Histogram of Screen Time",
+       xlab = "Screen Time (hrs)",
+       ylab = "Frequency",
+       border = "white",
+       breaks = 5)
>
> # -----
> # Q1b: Density Plot - Screen Time
> # -----
> new_window()
> print(
+   ggplot(app_data, aes(x = Screen_Time)) +
+     geom_density(fill = "orange", alpha = 0.6, color = "darkorange", lwd = 1) +
+     geom_vline(aes(xintercept = mean(Screen_Time)),
+                 color = "red", linetype = "dashed", lwd = 1) +
+     geom_vline(aes(xintercept = median(Screen_Time)),
+                 color = "blue", linetype = "dashed", lwd = 1) +
+     annotate("text", x = mean(app_data$Screen_Time) + 0.3,
+                 y = 0.18, label = "Mean", color = "red", size = 3.5) +
+     annotate("text", x = median(app_data$Screen_Time) - 0.3,
+                 y = 0.15, label = "Median", color = "blue", size = 3.5) +
+     labs(title = "Density Plot of Screen Time",
+          x = "Screen Time (hrs)",
+          y = "Density") +
+     theme_minimal()
+ )
>
> # -----
> # Q2: Scatter Plot - Data Used vs. Screen Time + Correlation + Trend Line
```

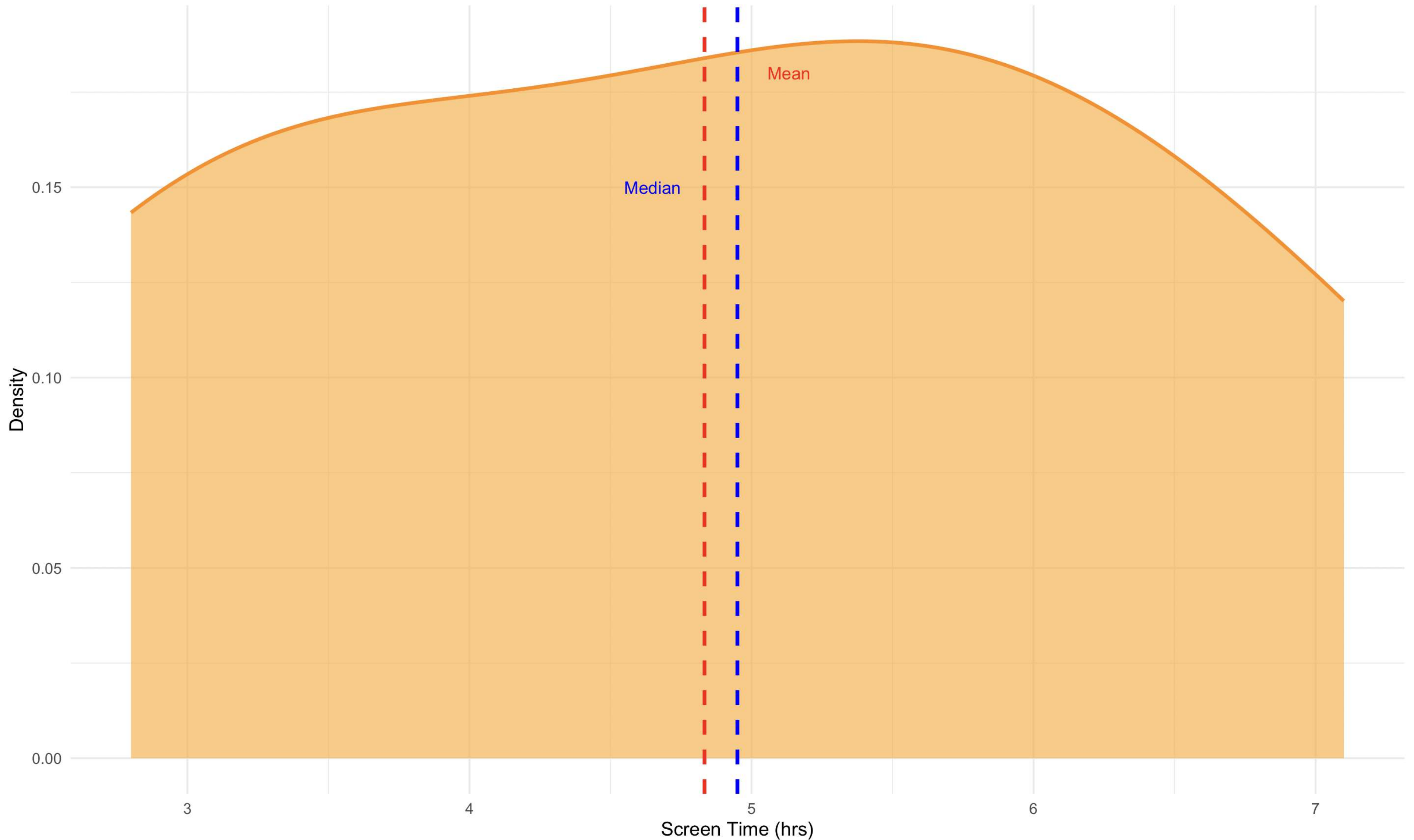


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+   annotate("text", x = median(app_data$Screen_time) - 0.3,
+       y = 0.15, label = "Median", color = "blue", size = 3.5) +
+   labs(title = "Density Plot of Screen Time",
+       x = "Screen Time (hrs)",
+       y = "Density") +
+   theme_minimal()
+ )
>
> # -----
> # Q2: Scatter Plot - Data Used vs Screen Time + Correlation + Trend Line
> # -----
> corr_val <- round(cor(app_data$Data_Used, app_data$Screen_Time), 3)
> cat("Correlation between Data_Used and Screen_Time:", corr_val, "\n")
Correlation between Data_Used and Screen_Time: 0.997
>
> new_window()
> print(
+   ggplot(app_data, aes(x = Screen_Time, y = Data_Used)) +
+     geom_point(size = 4, color = "purple") +
+     geom_smooth(method = "lm", se = FALSE, color = "red", linetype = "dashed") +
+     geom_text(aes(label = User_ID), vjust = -1, size = 3.5) +
+     annotate("text", x = 3, y = 4.2,
+       label = paste("r =", corr_val),
+       color = "darkred", size = 4, fontface = "bold") +
+     labs(title = "Data Used vs Screen Time with Trend Line",
+       x = "Screen Time (hrs)",
+       y = "Data Used (GB)") +
+     theme_minimal()
+ )
`geom_smooth()` using formula = 'y ~ x'
>
> # -----
> # Q3: Average Satisfaction by Gender - Bar Chart
> # -----
> avg_satisfaction <- aggregate(Satisfaction ~ Gender, data = app_data, FUN = mean)
> print(avg_satisfaction)
  Gender Satisfaction
1 Female    4.666667
2  Male    2.666667
>
> new_window()
> print(
+   ggplot(avg_satisfaction, aes(x = Gender, y = Satisfaction, fill = Gender)) +
+     geom_bar(stat = "identity", width = 0.5) +
+     geom_text(aes(label = round(Satisfaction, 2)),
+       vjust = -0.5, size = 4, fontface = "bold") +
+     scale_fill_manual(values = c("Female" = "pink", "Male" = "skyblue")) +
+     labs(title = "Average Satisfaction Score by Gender",
+       x = "Gender",
+       y = "Average Satisfaction Score") +
+     ylim(0, 6) +
+     theme_minimal()
+ )
>
```

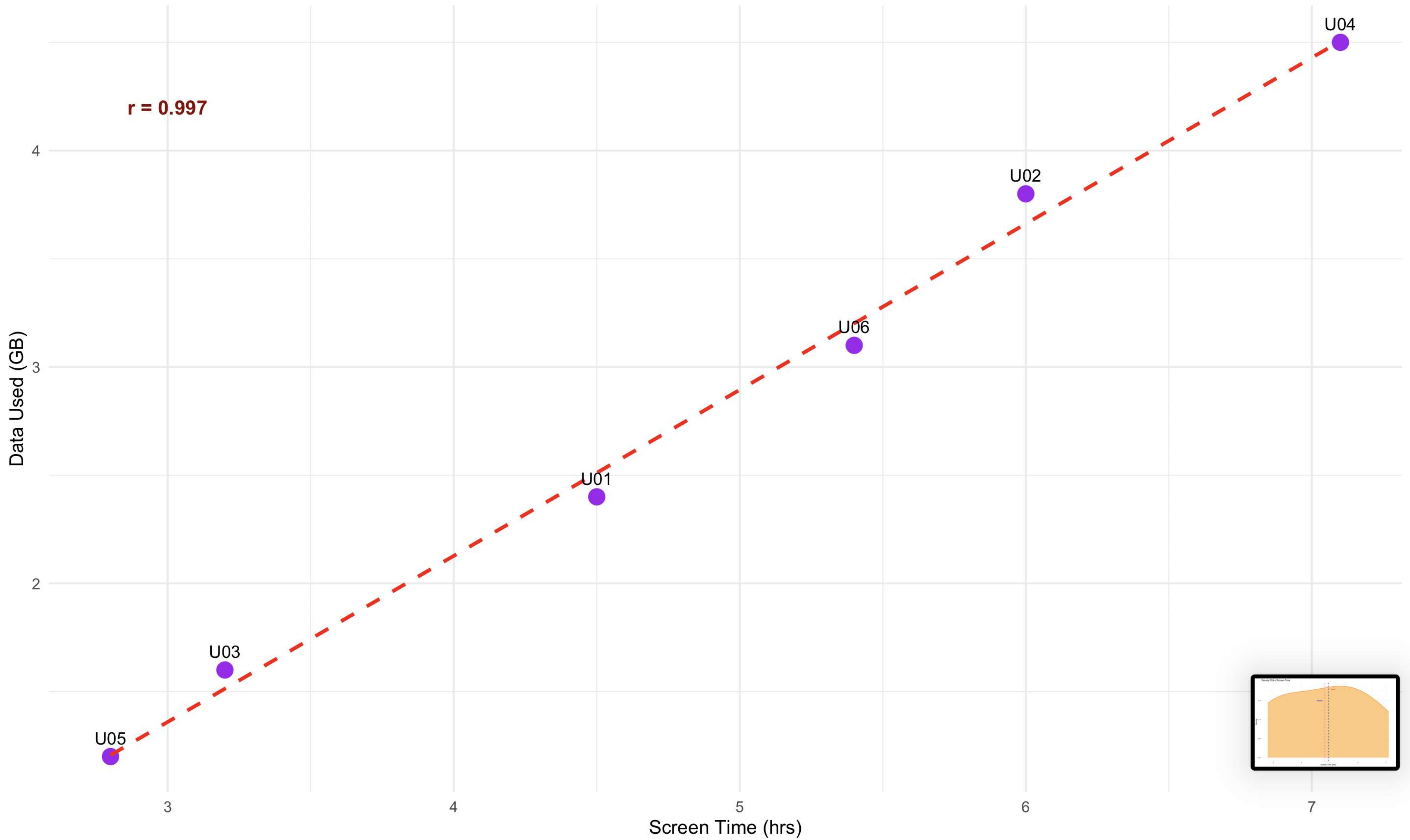
Histogram of Screen Time



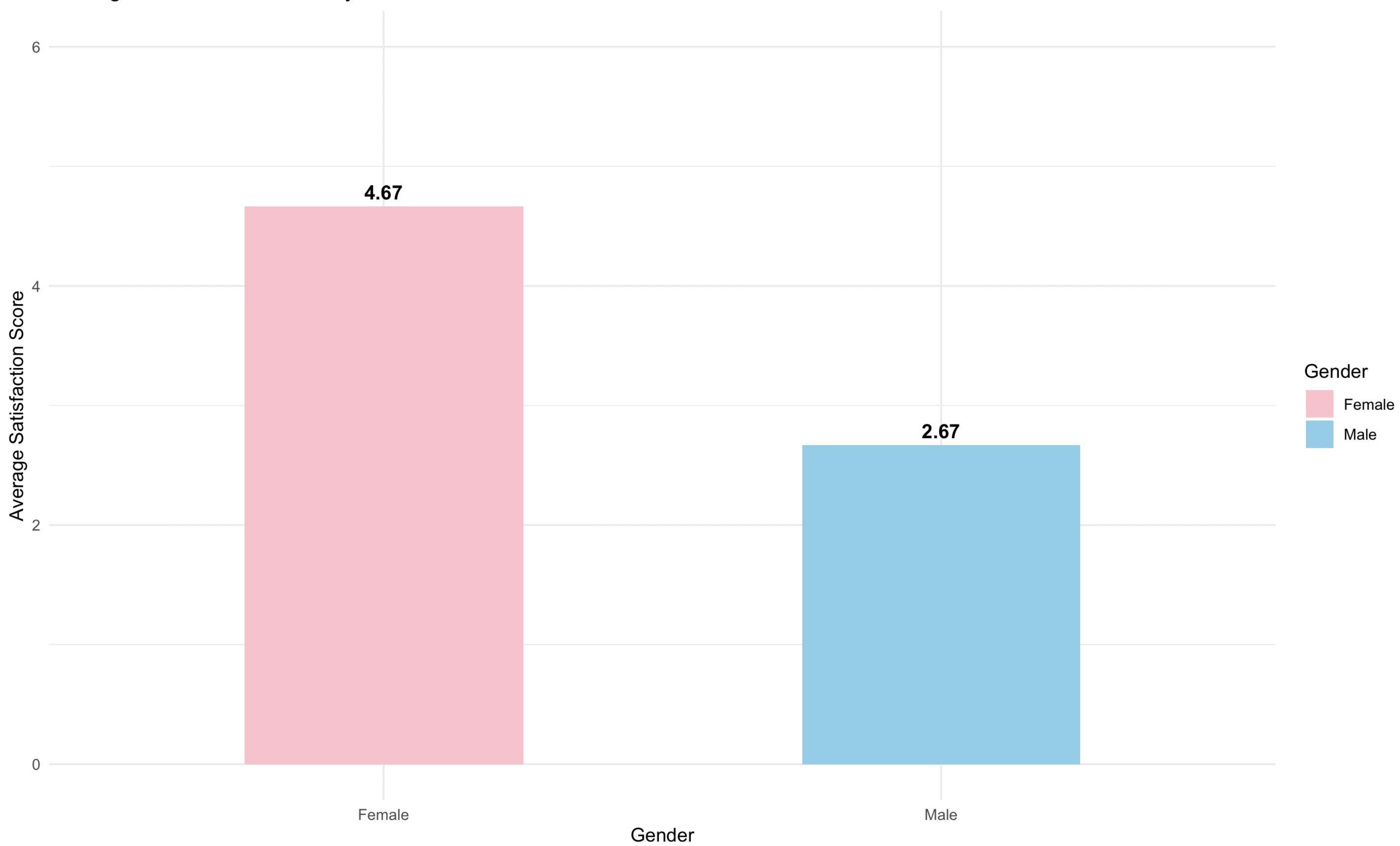
Density Plot of Screen Time

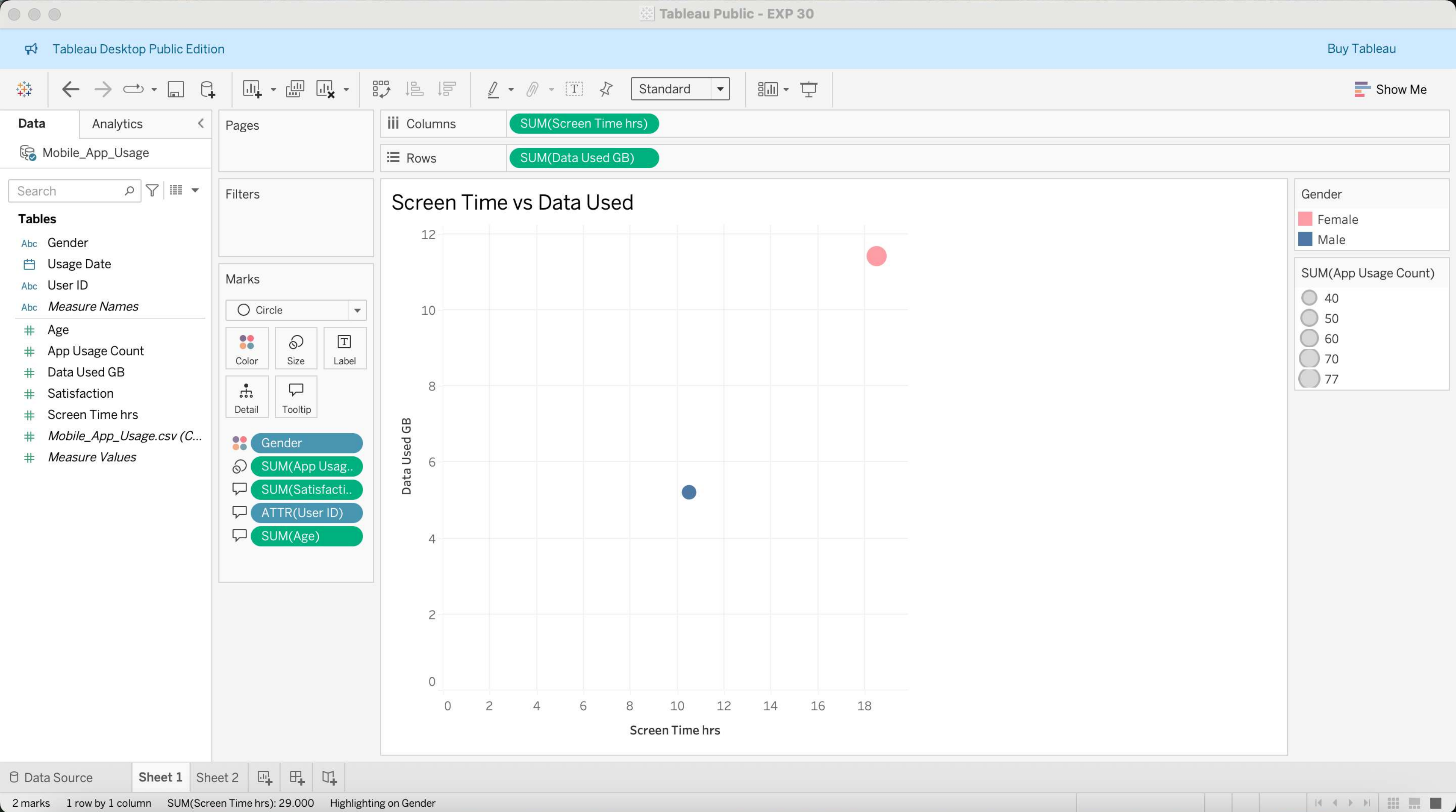


Data Used vs Screen Time with Trend Line



Average Satisfaction Score by Gender





Dashboard

Default

Phone

Device Preview

Size

Automatic

Sheets

Sheet 1

Sheet 2

Objects

Horizontal Container

Vertical Container

Text

Extension

Pulse Metric

Image

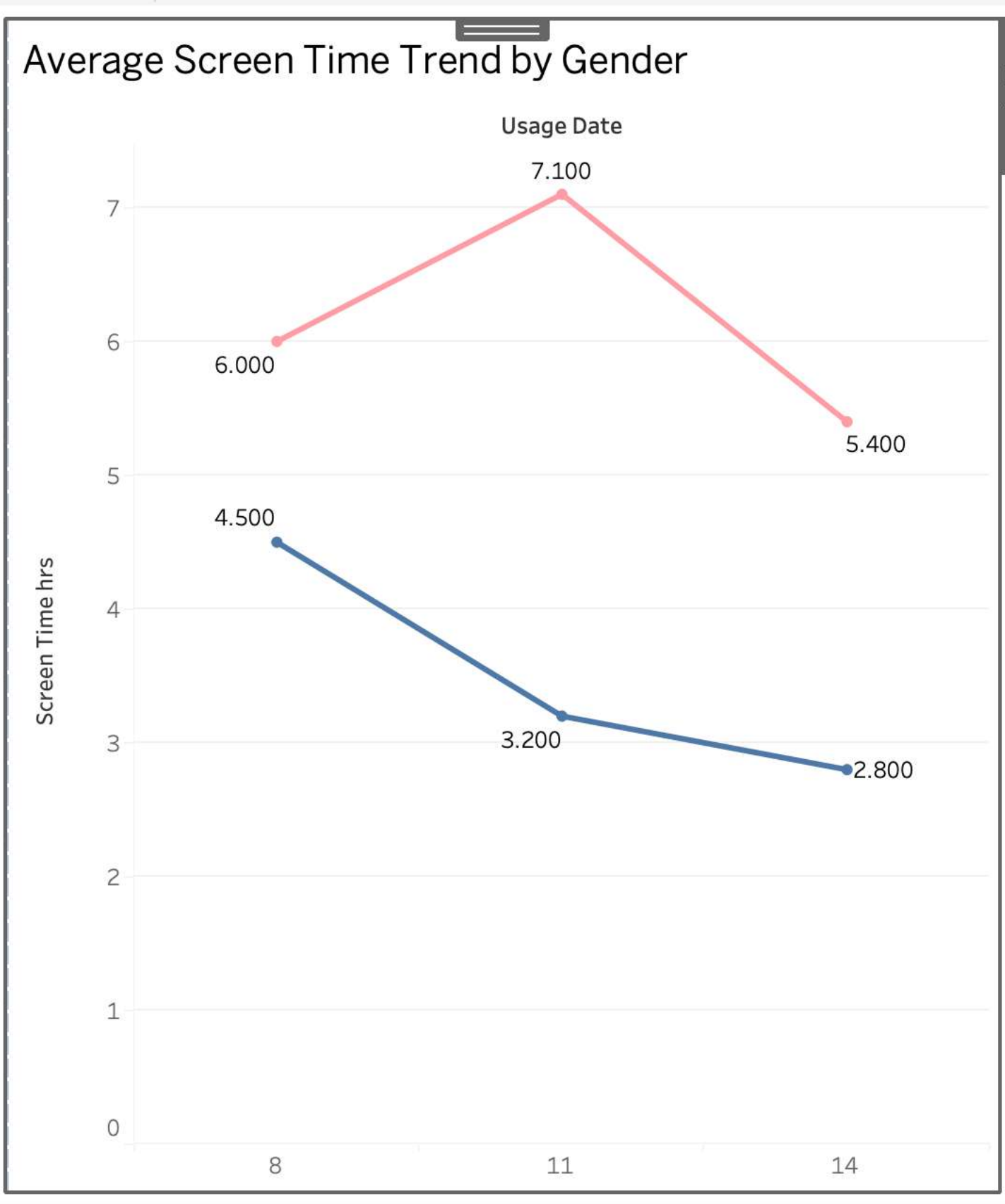
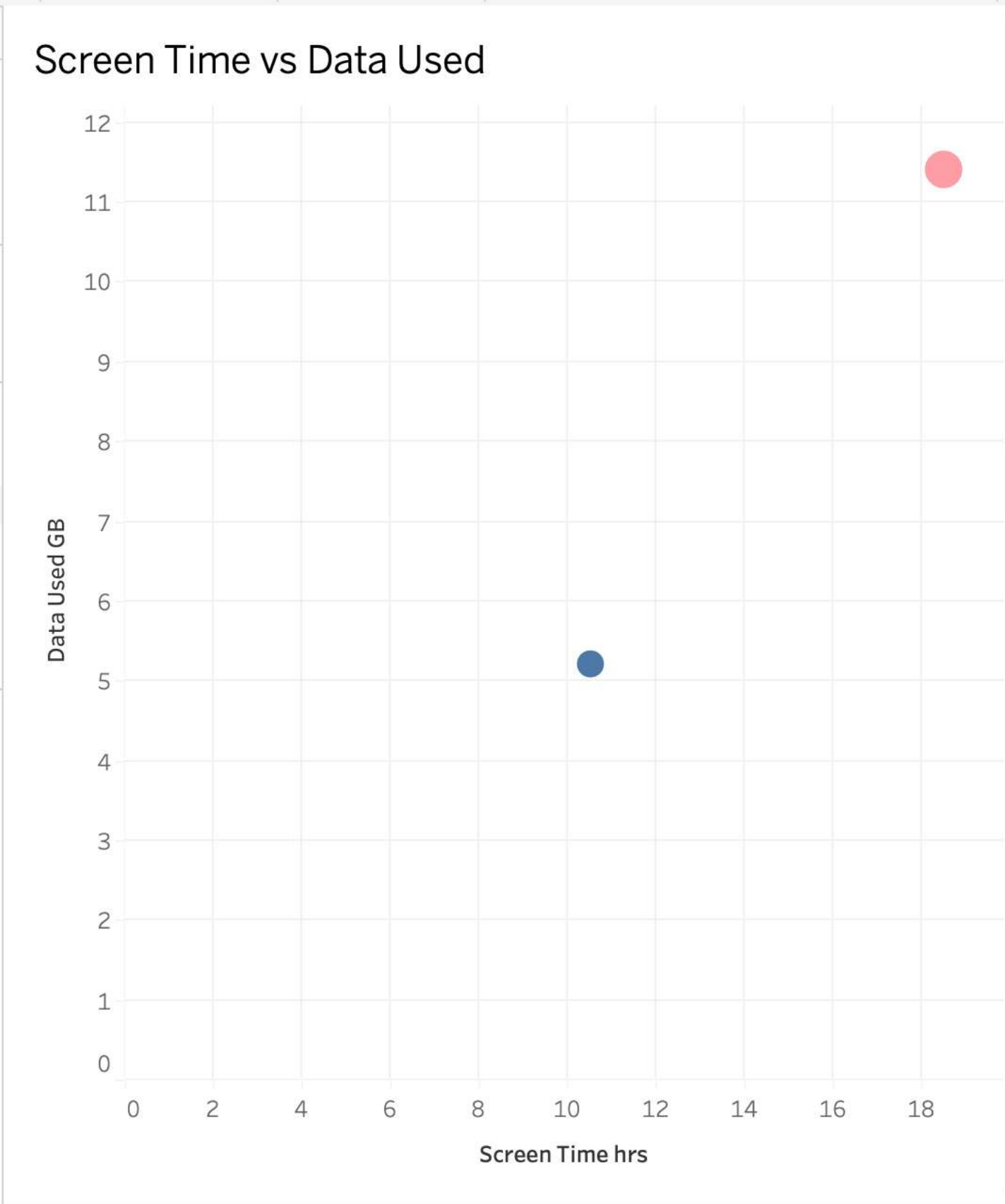
Blank

Workflow

Web Page

Tiled

Floating



Gender

Female

Male

Usage Count

40

50

60

70

77